

7. Let $f(x) = (x-a)(x-b)^2(x+c)$ with $0 < a < b < c$. Determine the solution for $f(x) < 0$. Express your answer(s) in interval notation. (For example, the interval (k, w) represents $k < x < w$.)
8. In a geometric sequence, the last term is 1458, the common ratio is -3 , and the sum of the terms is 1094. Determine the second term of this geometric sequence.
9. Let a , b , c , and d be positive real numbers. Let $a+b+c+d=10$, let $c=d$, and let $a^2+b^2+c^2+d^2=30$. The largest possible value of c can be expressed as $\frac{k+\sqrt{w}}{f}$ in simplified and reduced radical form where k , w , and f are positive integers. Determine the value of $(k+w+f)$.
10. Determine the numeric area of a triangle with sides 20 and 16 and the included angle between those two sides having measure 150° .
11. In $\triangle ABC$, $2\sin A + 3\cos B = 4$ and $3\sin B + 2\cos A = 3$. Determine the degree measure of $\angle C$.
12. Points $(1, 3, -1)$, $(-2, 1, 5)$, and $(3, 2, 6)$ lie on a plane. A standard equation for that plane can be written as $Ax + By + Cz + D = 0$ where A , B , C , and D are relatively prime integers and $A > 0$. Determine the sum $(A + B + C + D)$.
13. A parabola has the equation of $2x^2 + 12x = 3y$. The equation of the directrix of this parabola can be expressed in the form $x = k$ or $y = w$. Determine the value of k or w , whichever is appropriate, as an exact decimal. Express your answer as the equation $x = k$ or $y = w$.